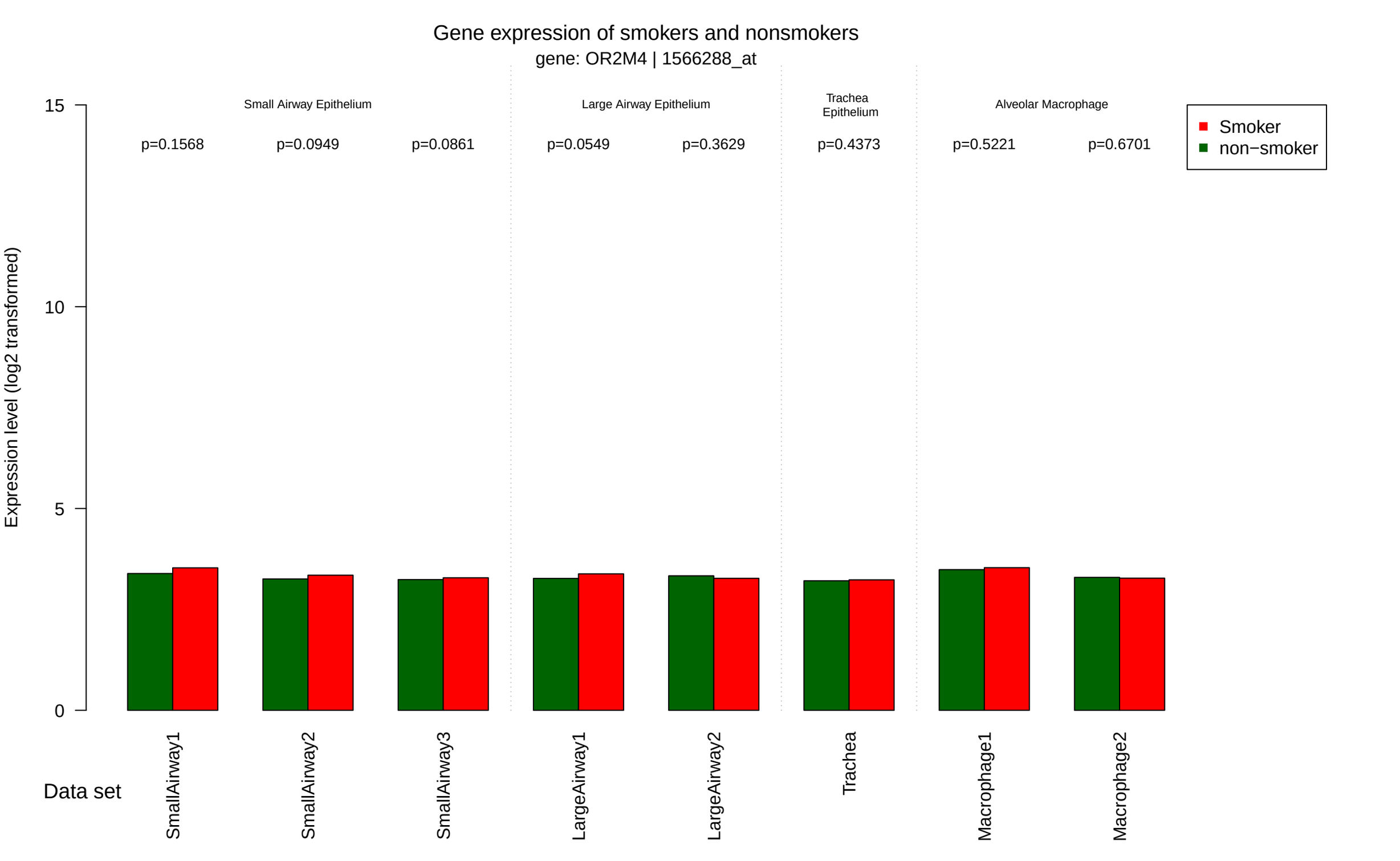
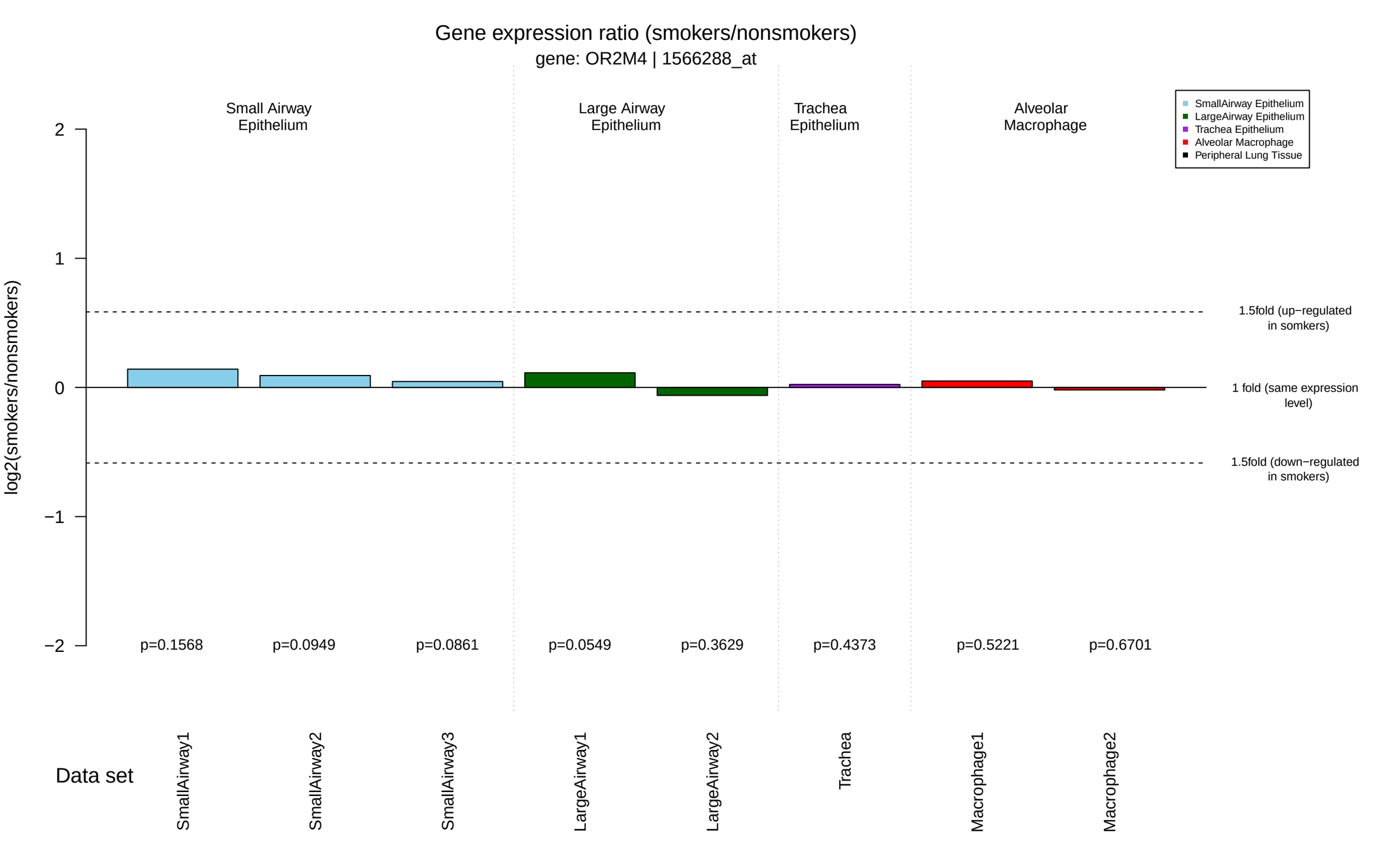
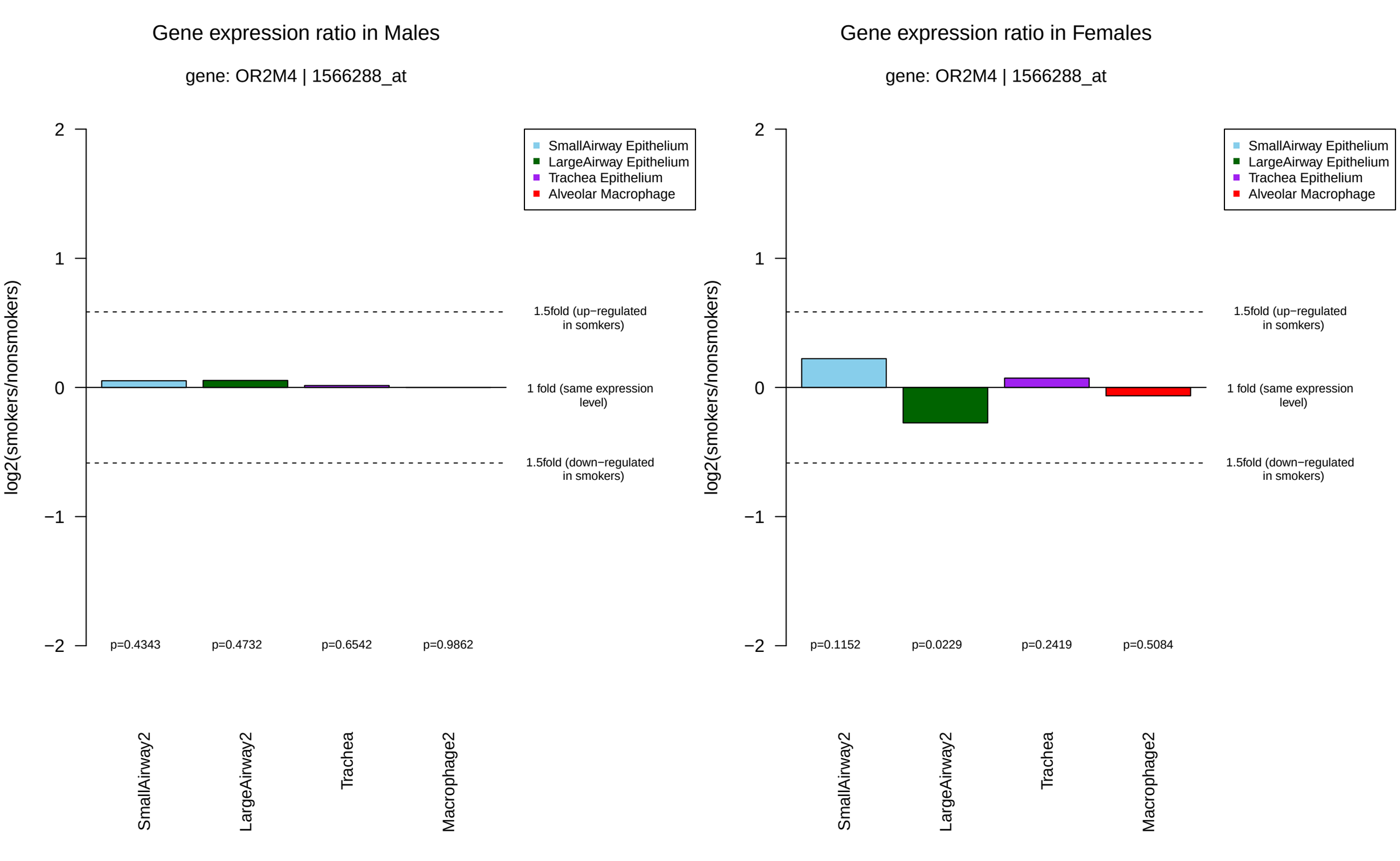
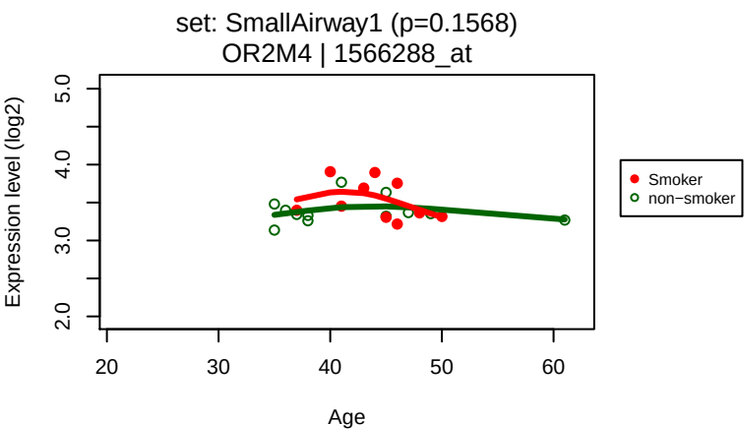
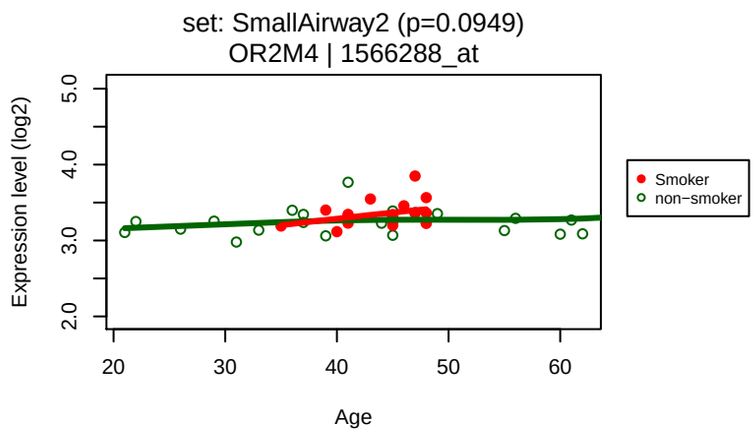
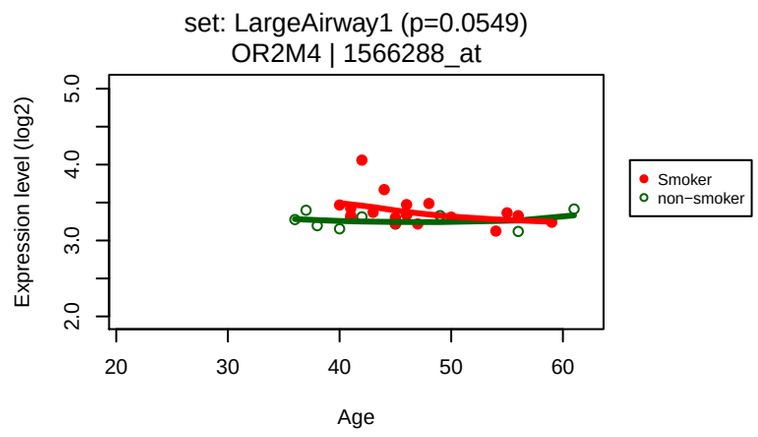

Gene expression of smokers and nonsmokers
gene: OR2M4 | 1566288_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1568
p=0.0949
p=0.0861
p=0.0549
p=0.3629
p=0.4373
p=0.5221
p=0.6701
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

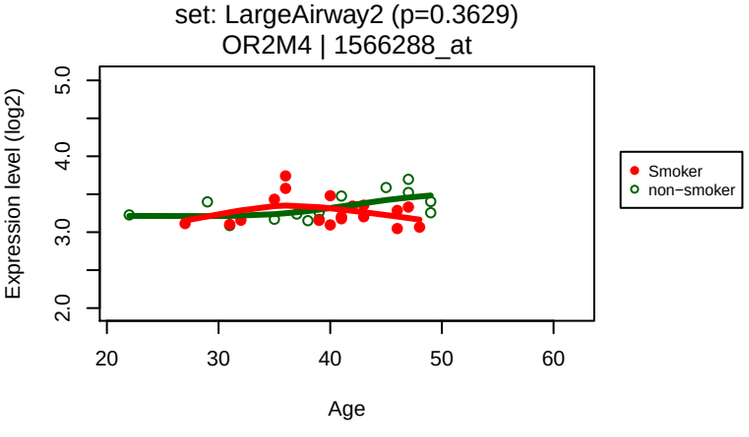

Gene expression ratio (smokers/nonsmokers)
gene: OR2M4 | 1566288_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1568
p=0.0949
p=0.0861
p=0.0549
p=0.3629
p=0.4373
p=0.5221
p=0.6701
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

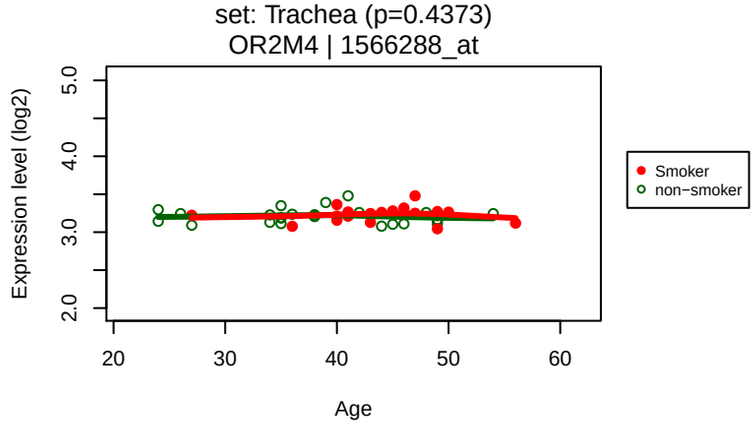

Gene expression ratio in Males
gene: OR2M4 | 1566288_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4343
p=0.4732
p=0.6542
p=0.9862
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR2M4 | 1566288_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1152
p=0.0229
p=0.2419
p=0.5084
SmallAirway2
LargeAirway2
Trachea
Macrophage2

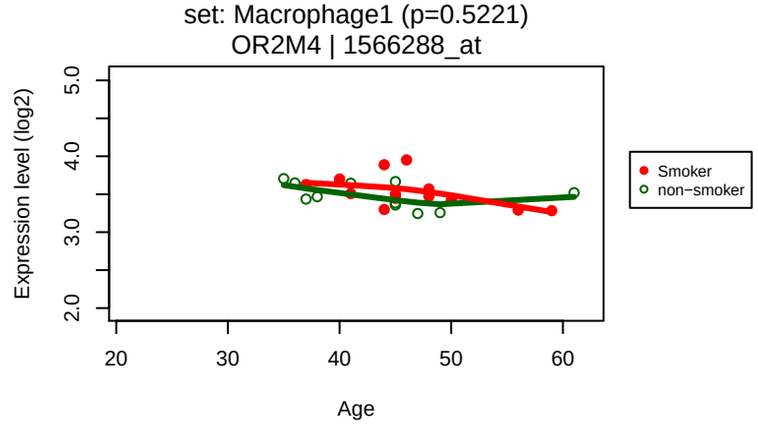

set: SmallAirway1 (p=0.1568)
OR2M4 | 1566288_at
Expression level (log2)
Age
Smoker
non−smoker

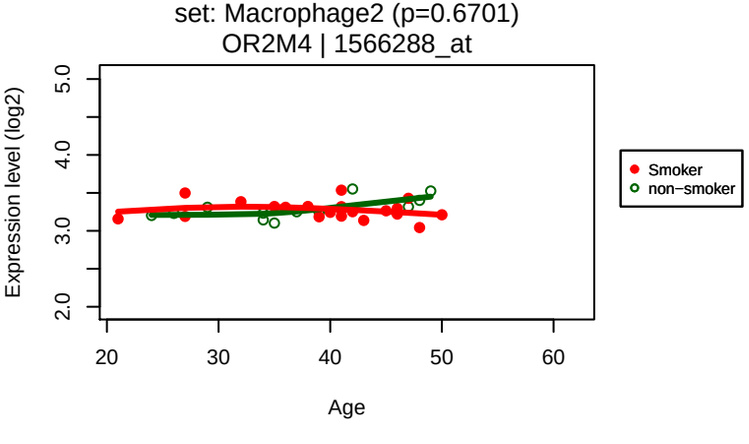

set: SmallAirway2 (p=0.0949)
OR2M4 | 1566288_at
Expression level (log2)
Age
Smoker
non−smoker

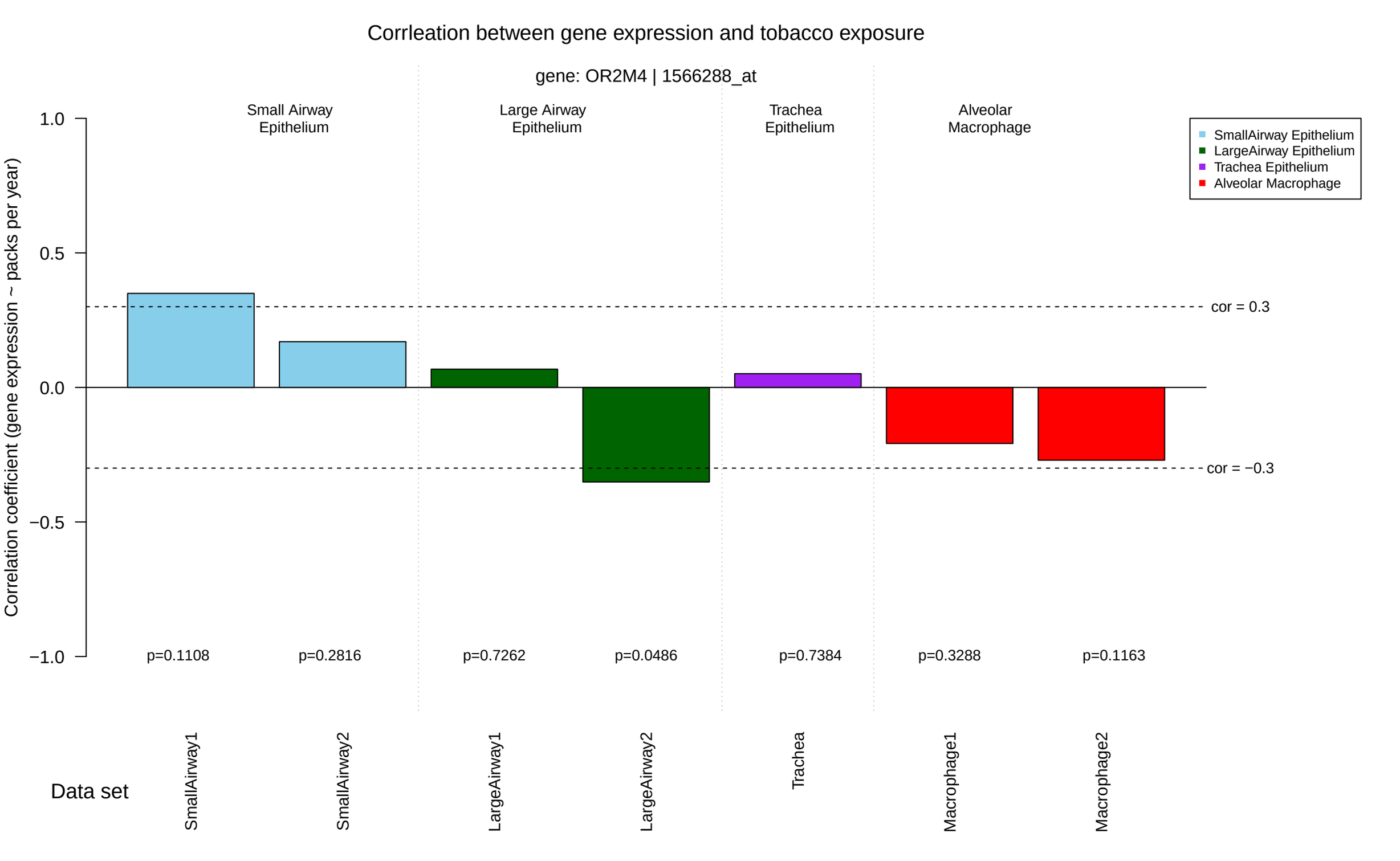

set: LargeAirway1 (p=0.0549)
OR2M4 | 1566288_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3629)
OR2M4 | 1566288_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4373)
OR2M4 | 1566288_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5221)
OR2M4 | 1566288_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6701)
OR2M4 | 1566288_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR2M4 | 1566288_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1108
p=0.2816
p=0.7262
p=0.0486
p=0.7384
p=0.3288
p=0.1163
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2